Cisco Packet Tracer - Basic Commands

Aim

To familiarize students with the **Cisco IOS command-line interface (CLI)** using Cisco Packet Tracer and to practice basic router monitoring, configuration, and interface configuration commands.

Procedure

Knowing the Current Mode

Step 1: Access the router CLI
Click:

Router → CLI

You will initially see:

Router>
The symbol > indicates **User EXEC mode**.

Step 2: Switch to Privileged EXEC mode
Enter:

Router> enable
Router#
The symbol # indicates **Privileged EXEC mode**.

Step 3: Return to User EXEC mode
Router# disable
Router>

Observation

Command	Result
enable	User EXEC → Privileged EXEC
disable	Privileged EXEC → User EXEC

Switching to Configuration Mode

From privileged EXEC mode:

Router# configure terminal
Router(config)#
or the abbreviated form:

Router# conf t
The prompt changes to:

Router(config)#
This indicates **Global Configuration mode**.

To return:

Router(config)# end
Router#

Obtaining Router Information
The command:

Router# show version
provides important information about the router.

Information obtained includes:

- Router model/type
- Cisco IOS version
- IOS image
- Configuration register
- Amount of RAM
- Flash memory
- Number and type of interfaces
- System uptime
- Serial number and hardware information
- Configuration register

Command
Router# show version

Important observation
Students should identify the following from the output:

Viewing Routing Protocol Information

Use:

Router# show ip protocols
This command displays information about the **IP routing protocols** configured on the router.

It can show:

- Routing protocol being used
- Networks advertised
- Routing protocol timers
- Administrative information
- Passive interfaces
- Sources of routing information

If no dynamic routing protocol has been configured, the router may indicate that no routing protocol is running.

Command
Router# show ip protocols

Observation
Routing protocol configured:

Networks advertised:

Viewing the Routing Table

Use:

Router# show ip route
The command displays the router's **IP routing table**.

Example:

Router# show ip route
Typical route codes include:

Code	Meaning
C	Connected
L	Local
S	Static
R	RIP
O	OSPF
D	EIGRP

For example:

C 192.168.1.0/24 is directly connected, GigabitEthernet0/0
L 192.168.1.1/32 is directly connected, GigabitEthernet0/0

Important point
The routing table tells the router **where and how to forward packets**.

Viewing Command History

Cisco IOS maintains a history of recently entered commands.

Use:

Router# show history
Example:

Router# show history
The command displays recently executed commands.

Viewing the Router Clock

Use:

Router# show clock
Example:

Router# show clock
The output displays the current time maintained by the router.

Viewing the List of Hosts

Cisco IOS can maintain a table that maps host names to IP addresses.

Use:

Router# show hosts
The output displays information about the configured or learned host names.

Example

Router# show hosts

Displaying Interface Statistics

There are two useful forms of interface information.

a. Detailed Interface Information

Use:

Router# show interfaces
This displays detailed information about all interfaces.

For a particular interface:

Router# show interfaces gigabitEthernet 0/0
or depending on the router:

Router# show interfaces serial 0/0/0

Information available includes:

- Interface status
- Line protocol status
- MAC address
- IP-related information
- MTU
- Bandwidth
- Delay
- Encapsulation
- Input packets
- Output packets
- Errors
- Collisions
- Interface counters

Example:

GigabitEthernet0/0 is up, line protocol is up
This indicates that both the physical interface and data-link protocol are operational.

b. Brief Interface Information

Use:

Router# show ip interface brief
This is one of the **most useful troubleshooting commands**.

Example:

```
Interface              IP-Address      OK? Method Status
GigabitEthernet0/0     192.168.1.1     YES manual up
GigabitEthernet0/1     unassigned      YES unset  administratively down
Serial0/0/0            10.0.0.1        YES manual up
```

Interpretation

Status	Meaning
up	Physical interface is operational
down	Physical/link problem
administratively down	Interface has been disabled using shutdown

The command:

show ip interface brief
provides a **quick summary**, whereas:

show interfaces
provides **detailed statistics**.

Determining DTE or DCE

Serial connections use two types of devices:

- **DTE — Data Terminal Equipment**
- **DCE — Data Communications Equipment**

The **DCE side provides clocking** for a synchronous serial connection.

Use:

Router# show controllers serial 0/0/0
Example output may contain information indicating whether the interface is:

DCE
or:

DTE

Important rule

Clock rate is configured on the DCE side.

For example:

Router0(config-if)# clock rate 64000

Configuring an Ethernet Interface

Suppose Router0's GigabitEthernet0/0 interface has to be configured with:

IP address: 192.168.1.1
Subnet mask: 255.255.255.0

Step 1: Enter privileged mode
Router> enable

Step 2: Enter global configuration mode
Router# configure terminal

Step 3: Select the interface
Router(config)# interface gigabitEthernet 0/0
The prompt becomes:

Router(config-if)#

Step 4: Assign the IP address
Router(config-if)# ip address 192.168.1.1 255.255.255.0

Step 5: Enable the interface
Router(config-if)# no shutdown
You may see messages such as:

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

Step 6: Exit
Router(config-if)# exit
Router(config)# exit
Router#

Step 7: Verify
Router# show ip interface brief
The interface should show:

GigabitEthernet0/0 192.168.1.1 YES manual up up

Configuring a Serial Interface

Assume Router0 is the **DCE** side.

Configure:

IP address: 10.0.0.1
Subnet mask: 255.255.255.252
Clock rate: 64000

Commands
Router# configure terminal

```
Router(config)# interface serial 0/0/0
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# end
```

Verify

```
Router# show ip interface brief
```

and:

```
Router# show controllers serial 0/0/0
```

Configuration of the Other Serial Router

Router1 is assumed to be the **DTE** side.

Configure:

IP address: 10.0.0.2
Subnet mask: 255.255.255.252

```
Router1# configure terminal
Router1(config)# interface serial 0/0/0
Router1(config-if)# ip address 10.0.0.2 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# end
```

Important

Do **not** configure:

```
clock rate 64000
```

on the DTE interface.

The clock is supplied by the DCE side.

Verification Commands

After completing the configuration, execute the following commands.

Check interface status

```
show ip interface brief
```

Check detailed interface information

```
show interfaces
```

Check routing table

```
show ip route
```

Check router information

```
show version
```

Check routing protocols

```
show ip protocols
```

Check clock
show clock

Check hosts
show hosts

Check command history
show history

Check DTE/DCE
show controllers serial 0/0/0

Save configuration
copy running-config startup-config

OUTPUT

Observation Table

Students can complete the following table in the lab record.

Sl. No.	Task	Command Used	Observation
1	Identify current mode	show ... / observe prompt	_____
2	Enter privileged mode	enable	_____
3	Enter configuration mode	configure terminal	_____
4	Obtain router information	show version	_____
5	View routing protocols	show ip protocols	_____
6	View routing table	show ip route	_____
7	View running configuration	show running-config	_____
8	Save configuration	copy run start	_____
9	View command history	show history	_____
10	View router clock	show clock	_____
11	View hosts	show hosts	_____
12	View detailed interface statistics	show interfaces	_____

Sl. No.	Task	Command Used	Observation
13	View brief interface status	show ip interface brief	_____
14	Determine DTE/DCE	show controllers serial 0/0/0	_____
15	Configure Ethernet interface	ip address ...	_____
16	Configure serial interface	ip address ...	_____
17	Enable interface	no shutdown	_____
18	Configure DCE clock	clock rate 64000	_____

Result

The basic Cisco IOS router commands were studied and practiced using Cisco Packet Tracer. The router information, routing information, interface statistics, configuration information, clock, host information, and command history were examined. Ethernet and serial interfaces were configured with appropriate IP addresses and subnet masks, and the interfaces were enabled and verified successfully.

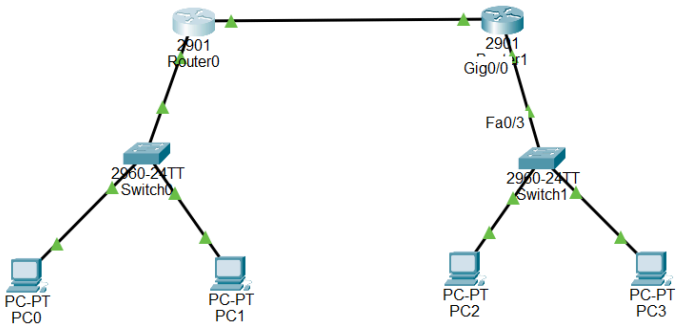
12. Set up static routing for the network . Once the routes are set up, display the routing table and verify the connectivity using ping.

Aim

To connect two separate LANs using **two routers** in Cisco Packet Tracer, configure IP addresses on the PCs and router interfaces, configure **static routes** to reach non-directly-connected networks, and verify end-to-end connectivity using the ping command.

Network Topology

Use the following topology:



Complete Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
Router0	G0/0	192.168.1.1	255.255.255.0	—
Router0	G0/1	10.0.0.1	255.255.255.252	—
Router1	G0/0	10.0.0.2	255.255.255.252	—
Router1	G0/1	192.168.2.1	255.255.255.0	—
PC2	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1

Networks

Network	Purpose
192.168.1.0/24	LAN 1
10.0.0.0/30	Router-to-router link

192.168.2.0/24 LAN 2

Devices Required

Device	Quantity
Cisco Router	2
Cisco 2960 Switch	2
PC	4
Copper Straight-Through Cable	6

Cable connections

- PC0 → Switch0
- PC1 → Switch0
- Switch0 → Router0
- Router0 → Router1
- Router1 → Switch1
- Switch1 → PC2
- Switch1 → PC3

Procedure

Step 1: Open Cisco Packet Tracer

Start Cisco Packet Tracer.

Create a new blank workspace.

Step 2: Place Two Routers

Select Network Devices → Routers.

Cisco 1941-type router can be used.

Step 3: Place Two Switches

Place two Cisco 2960 switches.

Switch0 → LAN 1

Switch1 → LAN 2

Step 4: Place Four PCs

Place:

PC0
PC1
PC2
PC3
Arrange them:

```
      LAN 1                    LAN 2

 PC0      PC1              PC2      PC3
  |       |                |       |
  +---- Switch0 ---- R1 ---- R2 ---- Switch1 ----+
```

Step 5: Connect LAN 1

Using **Copper Straight-Through** cables:

PC0 FastEthernet0 → Switch0 Fa0/1

PC1 FastEthernet0 → Switch0 Fa0/2

Switch0 Fa0/24 → Router0 G0/0
LAN 1 becomes:

```
 PC0
  |
  |
Switch0
  |
  |
 R1 G0/0
```

Step 6: Connect the Two Routers

Use a **Copper Cross-Over** cable for the router-to-router Ethernet connection if your Packet Tracer router/interface setup requires the traditional cable type.

Connect:

```
Router0 G0/1
     |
     | Cross-Over
```

|
Router1 G0/0

Step 7: Connect LAN 2

Using Copper Straight-Through cables:

Router1 G0/1 → Switch1 Fa0/24

Switch1 Fa0/1 → PC2 FastEthernet0

Switch1 Fa0/2 → PC3 FastEthernet0

Step 8: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 9: Configure PC1

Enter:

IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 10: Configure PC2

Enter:

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Step 11: Configure PC3

Enter:

IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Step12: Configure Router0 G0/0

Click **Router0 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to **LAN 1**.

Step 13: Configure Router0 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to Router1.

Step 14: Configure Router1 G0/0

Click **Router1 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

Task 15: Configure Router1 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Then:

```
Router(config)# end
```

Step 16: Verify Router0 Interfaces

On Router0:

show ip interface brief
Expected:

Interface	IP Address	Status	Protocol
G0/0	192.168.1.1	up	up
G0/1	10.0.0.1	up	up

Both should ideally show:

up/up

Step 17: Verify Router1 Interfaces

On Router1:

show ip interface brief
Expected:

Interface	IP Address	Status	Protocol
G0/0	10.0.0.2	up	up
G0/1	192.168.2.1	up	up

Step 18: Test the Router-to-Router Link

This is an important intermediate test.

From Router0:

ping 10.0.0.2
It should succeed.

From Router1:

ping 10.0.0.1
It should also succeed.

Step 19: Examine Router0's Routing Table

On Router0:

show ip route
Router0 should know about:

192.168.1.0/24
and:

10.0.0.0/30

Step 20: Examine Router1's Routing Table

On Router1:

show ip route
Router1 should know about:

10.0.0.0/30
and:

192.168.2.0/24
But it should **not yet know about**:

192.168.1.0/24

Step 21: Test LAN 1 Connectivity

From PC0:

ping 192.168.1.11
This should succeed because PC0 and PC1 are on the same LAN.

Step 22: Test LAN 2 Connectivity

From PC2:

ping 192.168.2.11
This should succeed because PC2 and PC3 are on the same LAN.

Step 23: Test Communication between the Two LANs

Now perform:

From PC0:

ping 192.168.2.10
This is an important test.

Expected result at this stage:
The ping should fail.

Configure Static Routes

Step 24: Configure a Static Route on Router0

On Router0:

Router0# configure terminal
Enter:

Router0(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
Then:

Router0(config)# end

Step 25: Configure a Static Route on Router1

On Router1:

Router1# configure terminal
Enter:

Router1(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1
Then:

Router1(config)# end

Step 26: Check Router0

Enter:

Router0# show ip route
You should now see an entry similar to:

S 192.168.2.0/24 [1/0] via 10.0.0.2
The letter:

S
means:

Static route

Step 27: Check Router1

Enter:

Router1# show ip route
You should see:

S 192.168.1.0/24 [1/0] via 10.0.0.1
Again:

S = Static

Step 28: Test Inter-LAN Connectivity Again

Now return to PC0.

Execute:

ping 192.168.2.10
This time the ping should succeed.

The packet travels:

PC0
|
| 192.168.1.10
↓
Switch0
|
↓
Router0
|
| 10.0.0.1 → 10.0.0.2
↓
Router1
|
↓
Switch1
|
↓
PC2
192.168.2.10

12. Task 29: Test the Reverse Direction

From PC2:

ping 192.168.1.10
This should also succeed.

The reverse path is:

PC2
↓
Switch1
↓
Router1
↓
Router0
↓

Switch0
↓
PC0
This demonstrates that **both routers require appropriate routing information**.

13. Connectivity Verification Table

Students should complete the following table.

Source	Destination	Command	Before Static Route	After Static Route
PC0	PC1	ping 192.168.1.11	Success	Success
PC2	PC3	ping 192.168.2.11	Success	Success
PC0	PC2	ping 192.168.2.10	Failure	Success
PC2	PC0	ping 192.168.1.10	Failure	Success
PC1	PC3	ping 192.168.2.11	Failure	Success
PC3	PC1	ping 192.168.1.11	Failure	Success

This table is particularly useful because it demonstrates **the effect of static routing** rather than merely asking students to type commands.

Step 31: Examine the Routing Table Again

On Router0:

show ip route
Students should identify:

C 192.168.1.0/24
C 10.0.0.0/30
S 192.168.2.0/24
On Router1:

show ip route
Students should identify:

C 10.0.0.0/30

C 192.168.2.0/24
S 192.168.1.0/24

20. Important Commands Used

Purpose	Command
Enter privileged mode	enable
Enter configuration mode	configure terminal
Configure interface	interface gigabitEthernet 0/0
Assign IP address	ip address IP MASK
Enable interface	no shutdown
Display interface status	show ip interface brief
Display routing table	show ip route
Add static route	ip route NETWORK MASK NEXT-HOP
Remove static route	no ip route NETWORK MASK NEXT-HOP
Test connectivity	ping IP-address
Trace packet path	tracert IP-address
Display configuration	show running-config
Save configuration	copy running-config startup-config

21. Static Route Commands — Most Important Part

Students should specifically record these two commands.

Router0
Router0(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Router1
Router1(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

22. Routing Table Observation

Students should complete this table.

Router0

Destination Network	Route Type	Next Hop	Interface
192.168.1.0/24	Connected	—	G0/0
10.0.0.0/30	Connected	—	G0/1
192.168.2.0/24	Static	10.0.0.2	G0/1

Router1

Destination Network	Route Type	Next Hop	Interface
10.0.0.0/30	Connected	—	G0/0
192.168.2.0/24	Connected	—	G0/1
192.168.1.0/24	Static	10.0.0.1	G0/0

.

23. Observation

PC Configuration

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	192.168.2.1

Router Configuration

Router	Interface	IP Address
Router0	G0/0	192.168.1.1
Router0	G0/1	10.0.0.1
Router1	G0/0	10.0.0.2

Router1 G0/1 192.168.2.1

Result

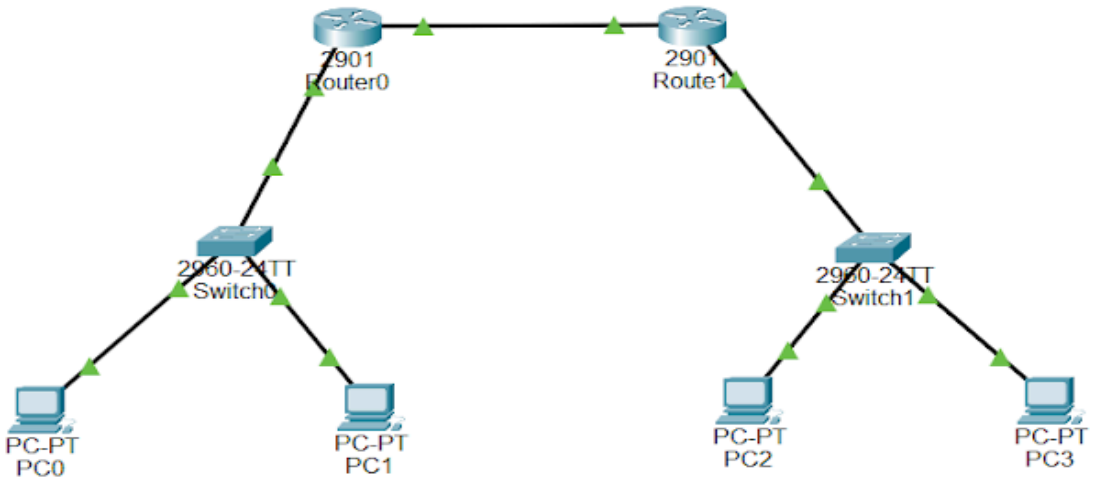
Two separate LANs were successfully connected using two routers in Cisco Packet Tracer. The router interfaces, PCs, subnet masks, and default gateways were configured. Static routes were configured on both routers to provide paths to the non-directly-connected LANs. Connectivity between the two LANs was successfully verified using ping.

13. Implement RIPv2 routing for the network

Aim

To configure dynamic routing using Routing Information Protocol (RIP Version 2) between two routers in Cisco Packet Tracer and verify communication between two different LANs.

Network Topology



Complete Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
Router0	G0/0	192.168.1.1	255.255.255.0	—
Router0	G0/1	10.0.0.1	255.255.255.252	—
Router1	G0/0	10.0.0.2	255.255.255.252	—
Router1	G0/1	192.168.2.1	255.255.255.0	—
PC2	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1

Networks

Network	Purpose
192.168.1.0/24	LAN 1
10.0.0.0/30	Router-to-router link

192.168.2.0/24 LAN 2

Devices Required

Device	**Quantity**
Cisco Router	2
Cisco 2960 Switch	2
PC	4
Copper Straight-Through Cable	6
Router-to-router Ethernet cable	1

Cable connections

- PC0 → Switch0
- PC1 → Switch0
- Switch0 → Router0
- Router0 → Router1
- Router1 → Switch1
- Switch1 → PC2
- Switch1 → PC3

Procedure

Step 1: Open Cisco Packet Tracer

Start Cisco Packet Tracer.

Create a new blank workspace.

Step 2: Place Two Routers

Select Network Devices → Routers.

Cisco 1941-type router can be used.

Step 3: Place Two Switches

Place two Cisco 2960 switches.

Switch0 → LAN 1
Switch1 → LAN 2

Step 4: Place Four PCs

Place:

PC0
PC1
PC2
PC3
Arrange them:

```
      LAN 1                    LAN 2

 PC0      PC1               PC2      PC3
  |       |                 |        |
  +---- Switch0 ---- R1 ---- R2 ---- Switch1 ----+
```

Step 5: Connect LAN 1

Using **Copper Straight-Through** cables:

PC0 FastEthernet0 → Switch0 Fa0/1

PC1 FastEthernet0 → Switch0 Fa0/2

Switch0 Fa0/24 → Router0 G0/0
LAN 1 becomes:

```
 PC0
  |
  |
Switch0
  |
  |
 R1 G0/0
```

Step 6: Connect the Two Routers

Use a **Copper Cross-Over** cable for the router-to-router Ethernet connection if your Packet Tracer router/interface setup requires the traditional cable type.

Connect:

```
Router0 G0/1
     |
     | Cross-Over
     |
```

Router1 G0/0

Step 7: Connect LAN 2

Using Copper Straight-Through cables:

Router1 G0/1 → Switch1 Fa0/24

Switch1 Fa0/1 → PC2 FastEthernet0

Switch1 Fa0/2 → PC3 FastEthernet0

Step 8: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 9: Configure PC1

Enter:

IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 10: Configure PC2

Enter:

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Step 11: Configure PC3

Enter:

IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Step12: Configure Router0 G0/0

Click **Router 0 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to **LAN 1**.

Step 13: Configure Router0 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to Router1.

Step 14: Configure Router1 G0/0

Click **Router1 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

Task 15: Configure Router1 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Then:

```
Router(config)# end
```

Step 16: Verify Router0 Interfaces

On Router0:

show ip interface brief
Expected:

Interface	IP Address	Status	Protocol
G0/0	192.168.1.1	up	up
G0/1	10.0.0.1	up	up

Both should ideally show:

up/up

Step 17: Verify Router1 Interfaces

On Router1:

show ip interface brief
Expected:

Interface	IP Address	Status	Protocol
G0/0	10.0.0.2	up	up
G0/1	192.168.2.1	up	up

Step 18: Test the Router-to-Router Link

This is an important intermediate test.

From Router0:

ping 10.0.0.2
It should succeed.

From Router1:

ping 10.0.0.1
It should also succeed.

Step 19: Examine Router0's Routing Table

On Router0:

show ip route

Router0 should know about:

192.168.1.0/24
and:

10.0.0.0/30

Step 20: Examine Router1's Routing Table

On Router1:

show ip route
Router1 should know about:

10.0.0.0/30
and:

192.168.2.0/24
But it should **not yet know about**:

192.168.1.0/24

Step 21: Test LAN 1 Connectivity

From PC0:

ping 192.168.1.11
This should succeed because PC0 and PC1 are on the same LAN.

Step 22: Test LAN 2 Connectivity

From PC2:

ping 192.168.2.11
This should succeed because PC2 and PC3 are on the same LAN.

Step 23: Test Communication between the Two LANs

Now perform:

From PC0:

ping 192.168.2.10
This is an important test.

Expected result at this stage:
The ping should fail.

Step 24: Enable RIPv2 on Router0

Router0> enable
Router0# configure terminal
Router0(config)# router rip

```
Router0(config-router)# version 2
Router0(config-router)# no auto-summary
Router0(config-router)# network 192.168.1.0
Router0(config-router)# network 10.0.0.0
Router0(config-router)# end
```

Step 25: Enable RIPv2 on Router1

```
Router1> enable
Router1# configure terminal
Router1(config)# router rip
Router1(config-router)# version 2
Router1(config-router)# no auto-summary
Router1(config-router)# network 10.0.0.0
Router1(config-router)# network 192.168.2.0
Router1(config-router)# end
```

Router0 advertises:
192.168.1.0/24
and:
10.0.0.0/30
Router1 advertises:
192.168.2.0/24
and:
10.0.0.0/30
The routers exchange routing information through the:
10.0.0.0/30
network.
Router0 eventually learns:
192.168.2.0/24
through Router1.
Router1 eventually learns:
192.168.1.0/24
through Router0.
No ip route command was entered.

That is the key difference from static routing.

Step 26: Verify RIP Configuration

On Router0:
show ip protocols
Look for information indicating that:
Routing Protocol is "rip"
and that:
version 2 is being used.
Also observe the networks participating in RIP.
Repeat on Router1:
show ip protocols

Step 27: Check Router0's Routing Table

Wait a few seconds for RIP to exchange routing information.
Then execute:
show ip route
Router0 should now have a route similar to:
R 192.168.2.0/24 [120/1] via 10.0.0.2
The exact formatting can vary slightly by IOS/Packet Tracer version.
The important part is:
R
R means:
RIP-learned route

Step 28: Check Router1's Routing Table

On Router1:
show ip route
You should find something similar to:
R 192.168.1.0/24 [120/1] via 10.0.0.1
Again:
R = route learned through RIP

Step 29: Compare the Routing Tables

Before RIP
Router0:

C 192.168.1.0/24
C 10.0.0.0/30
Router1:

C 10.0.0.0/30
C 192.168.2.0/24
After RIP
Router0:

C 192.168.1.0/24
C 10.0.0.0/30
R 192.168.2.0/24
Router1:

C 10.0.0.0/30
C 192.168.2.0/24
R 192.168.1.0/24

Step 30: Connectivity Verification Table

Source	Destination	Command	Expected Result
PC0	PC1	ping 192.168.1.11	Success
PC2	PC3	ping 192.168.2.11	Success
PC0	PC2	ping 192.168.2.10	Success
PC2	PC0	ping 192.168.1.10	Success
PC1	PC3	ping 192.168.2.11	Success
PC3	PC1	ping 192.168.1.11	Success

Step 31: Examine the RIP Information

On Router0:
show ip protocols
Students should record:

- Routing protocol: _____
- RIP version: _____
- Networks participating in RIP: _____
- RIP timers, if displayed: _____

Repeat on Router1.

Step 32: Examine RIP Routes

Use:
show ip route
Students should identify routes beginning with:
R
Record:

Router	RIP-learned Network	Next Hop
Router0	_____	_____
Router1	_____	_____

Expected:

Router	RIP-learned Network	Next Hop
Router0	192.168.2.0/24	10.0.0.2
Router1	192.168.1.0/24	10.0.0.1

Step 33: Understand the RIP Metric

RIP uses hop count as its routing metric.
In this experiment:
Router0 → Router1 → LAN2
The LAN2 network is one router hop away from Router0.
Therefore, the route may appear as:
[120/1]
where:
120 = RIP administrative distance

1 = RIP metric / hop count

Step 34: Observation Table

PC Configuration

PC	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	192.168.2.1

Router Configuration

Router	Interface	IP Address
Router0	G0/0	192.168.1.1
Router0	G0/1	10.0.0.1
Router1	G0/0	10.0.0.2
Router1	G0/1	192.168.2.1

RIP Routing Table Observation

Router0

Destination	Route Type	Next Hop
192.168.1.0/24	Connected	—
10.0.0.0/30	Connected	—
192.168.2.0/24	RIP	10.0.0.2

Router1

Destination	Route Type	Next Hop
10.0.0.0/30	Connected	—
192.168.2.0/24	Connected	—
192.168.1.0/24	RIP	10.0.0.1

Result

Two separate LANs were successfully connected using two routers in Cisco Packet Tracer. RIPv2 was configured on both routers to dynamically exchange routing information. The routers automatically learned the remote LAN networks, and the dynamically learned routes were verified using the show ip route and show ip protocols commands. End-to-end connectivity between the two LANs was successfully established and verified using the ping command.

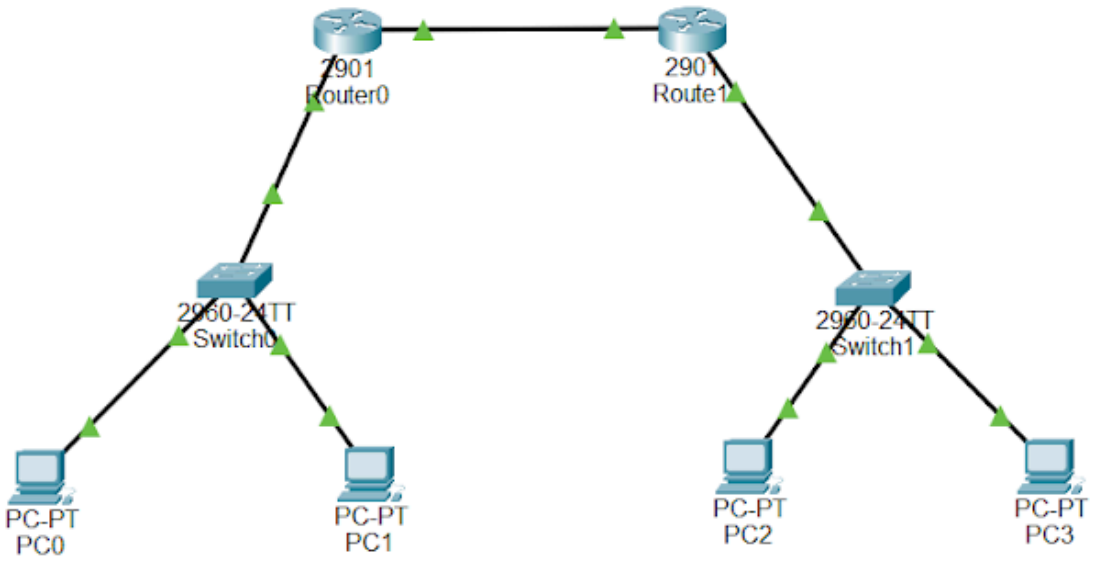
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14. Implement OSPF routing for the given network .Once the routes are set up, display the routing table and verify the connectivity using ping.

Aim

To configure Open Shortest Path First (OSPF) dynamic routing between two routers in Cisco Packet Tracer and verify automatic exchange of routing information and communication between two different LANs.

Network Topology



Complete Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1
Router0	G0/0	192.168.1.1	255.255.255.0	—
Router0	G0/1	10.0.0.1	255.255.255.252	—
Router1	G0/0	10.0.0.2	255.255.255.252	—
Router1	G0/1	192.168.2.1	255.255.255.0	—
PC2	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1

Devices Required

Device	Quantity
Cisco Router	2
Cisco 2960 Switch	2
PC	4
Copper Straight-Through Cable	6
Router-to-router Ethernet cable	1

Cable connections

LAN 1:

- PC0 → Switch0
- PC1 → Switch0
- Switch0 → Router0

Router connection:

- Router0 → Router1

LAN 2:

- Router1 → Switch1
- Switch1 → PC2
- Switch1 → PC3

Procedure

Step 1: Open Cisco Packet Tracer

Start Cisco Packet Tracer.

Create a new blank workspace.

Step 2: Place Two Routers

Select Network Devices → Routers.

Cisco 1941-type router can be used.

Step 3: Place Two Switches

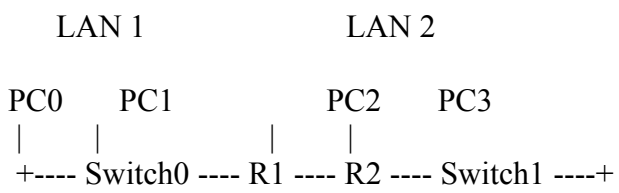
Place two Cisco 2960 switches.

Switch0 → LAN 1
Switch1 → LAN 2

Step 4: Place Four PCs

Place:

PC0
PC1
PC2
PC3
Arrange them:



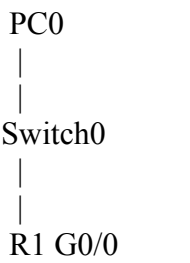
Step 5: Connect LAN 1

Using **Copper Straight-Through** cables:

PC0 FastEthernet0 → Switch0 Fa0/1

PC1 FastEthernet0 → Switch0 Fa0/2

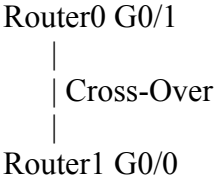
Switch0 Fa0/24 → Router0 G0/0
LAN 1 becomes:



Step 6: Connect the Two Routers

Use a **Copper Cross-Over** cable for the router-to-router Ethernet connection if your Packet Tracer router/interface setup requires the traditional cable type.

Connect:



Step 7: Connect LAN 2

Using Copper Straight-Through cables:

Router1 G0/1 → Switch1 Fa0/24

Switch1 Fa0/1 → PC2 FastEthernet0

Switch1 Fa0/2 → PC3 FastEthernet0

Step 8: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 9: Configure PC1

Enter:

IP Address: 192.168.1.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Step 10: Configure PC2

Enter:

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Step 11: Configure PC3

Enter:

IP Address: 192.168.2.11
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Step12: Configure Router0 G0/0

Click **Router 0 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to **LAN 1**.

Step 13: Configure Router0 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

This interface connects Router0 to Router1.

Step 14: Configure Router1 G0/0

Click **Router1 → CLI**.

Enter:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 10.0.0.2 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
```

Task 15: Configure Router1 G0/1

Enter:

```
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Then:

```
Router(config)# end
```

Step 16: Verify Router0 Interfaces

On Router0:

```
show ip interface brief
```

Expected:

Interface	IP Address	Status	Protocol
G0/0	192.168.1.1	up	up
G0/1	10.0.0.1	up	up

Both should ideally show:

up/up

Step 17: Verify Router1 Interfaces

On Router1:

show ip interface brief
Expected:

Interface	IP Address	Status	Protocol
G0/0	10.0.0.2	up	up
G0/1	192.168.2.1	up	up

Step 18: Test PC0 → PC1

From PC0:
ping 192.168.1.11
Expected:
Successful

Step 19: Test PC2 → PC3

From PC2:
ping 192.168.2.11
Expected:
Successful

Step 20: Test PC0 → PC2

From PC0:
ping 192.168.2.10
Expected:
Failure
Why?
Router0 does not yet know how to reach:
192.168.2.0/24
This gives students a clear reason for configuring OSPF.

Step 21: Check Router0

Enter:
show ip route
Router0 should have routes for:
192.168.1.0/24
10.0.0.0/30
but not:
192.168.2.0/24

Step 22: Check Router1

Enter:
show ip route
Router1 should have routes for:
10.0.0.0/30
192.168.2.0/24
but not:
192.168.1.0/24

Step 23: Start OSPF

Router0# configure terminal
Router0(config)# router ospf 1
Router0(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router0(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router0(config-router)# end

Here:
1
is the **OSPF process ID**.
It is locally significant to the router.

Repeat in Router1

Router1# configure terminal
Router1(config)# router ospf 1
Router1(config-router)# network 10.0.0.0 0.0.0.3 area 0
Router1(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router1(config-router)# end

Step 24: Understand the OSPF Network Command

In 192.168.1.0 0.0.0.255 area 0
The second value is a **wildcard mask**, not a subnet mask.
Subnet mask
255.255.255.0
Wildcard mask
0.0.0.255
The wildcard mask is essentially the inverse of the subnet mask.
Similarly:

255.255.255.252
has the wildcard mask:
0.0.0.3
Therefore:
network 10.0.0.0 0.0.0.3 area 0 matches the interfaces belonging to the 10.0.0.0/30 network.

Step 25: Verify OSPF Neighbor Relationship

This is one of the **most important commands in this experiment**.
On Router0:
show ip ospf neighbor
You should see Router1 listed as an OSPF neighbor.
Similarly, on Router1:
show ip ospf neighbor
Router0 should appear as a neighbor.
The exact output varies with the router model and Packet Tracer version, but the neighbor should reach a stable state such as:
FULL
Record:

Parameter	Observation
Neighbor Router ID	_____
Neighbor State	_____
Neighbor IP Address	_____
Local Interface	_____

The neighbor state should normally reach:

FULL

Important concept
An OSPF neighbor relationship means that the two routers have successfully established an OSPF relationship and can exchange routing information.

Step 26: Verify OSPF Configuration

On Router0:
show ip protocols
Students should identify:
Routing Protocol is "ospf 1"
and the networks participating in OSPF.
Repeat on Router1:
show ip protocols

Step 27: Examine the Routing Table After OSPF

On Router0:
show ip route
Router0 should now have an entry similar to:

O 192.168.2.0/24 [110/2] via 10.0.0.2
The exact cost can vary depending on the interfaces/router model.
The important observation is:
O
O means:OSPF-learned route

On Router1:
show ip route
You should find a route similar to:
O 192.168.1.0/24 [110/2] via 10.0.0.1
Again:
O = OSPF

Step 28: Compare the Routing Tables

Before OSPF
Router0:

C 192.168.1.0/24
C 10.0.0.0/30
Router1:

C 10.0.0.0/30
C 192.168.2.0/24
After OSPF
Router0:

C 192.168.1.0/24
C 10.0.0.0/30
O 192.168.2.0/24
Router1:

C 10.0.0.0/30
C 192.168.2.0/24
O 192.168.1.0/24

Step 29: Connectivity Verification Table

Source	Destination	Command	Expected Result
PC0	PC1	ping 192.168.1.11	Success
PC2	PC3	ping 192.168.2.11	Success
PC0	PC2	ping 192.168.2.10	Success

Source	Destination	Command	Expected Result
PC2	PC0	ping 192.168.1.10	Success
PC1	PC3	ping 192.168.2.11	Success
PC3	PC1	ping 192.168.1.11	Success

Result

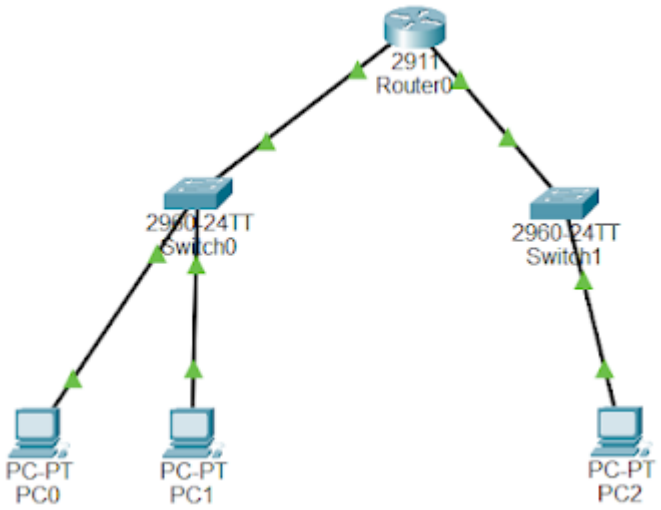
Two separate LANs were successfully connected using two routers in Cisco Packet Tracer. OSPF dynamic routing was configured using Area 0, and an OSPF neighbor relationship was successfully established between the two routers. The routers dynamically learned the remote LAN networks, which were verified using the show ip route and show ip ospf neighbour commands. End-to-end connectivity between the two LANs was successfully verified using the ping command

15. Configuring a Standard ACL to Restrict Access to a Specific IP Address

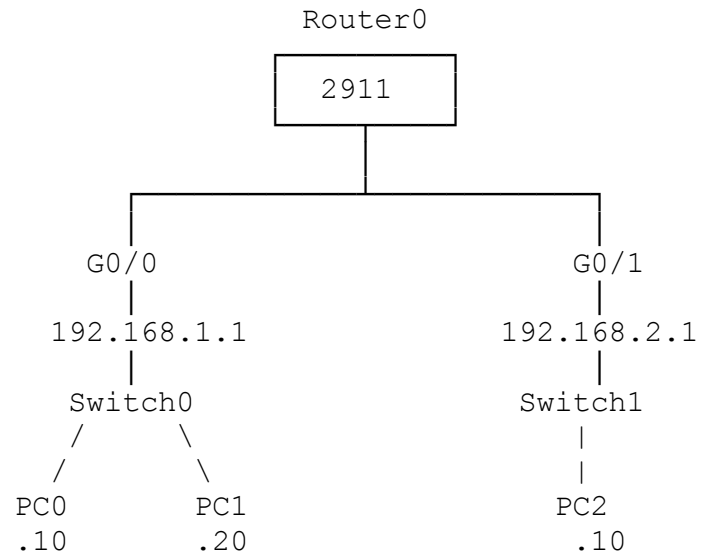
Aim

To configure a standard IPv4 Access Control List (ACL) on a Cisco router using Packet Tracer to restrict a specific IP address from accessing a destination network

Network Topology



We will use the following simple topology:



We want to achieve:

PC0 ——————————→ PC2 ALLOW
PC1 ——————————→ PC2 DENY

IP Addressing Scheme

We will use two IPv4 networks.

LAN 1
192.168.1.0/24

LAN 2
192.168.2.0/24

Device	Interface	IP Address	Subnet Mask	Gateway
Router0	G0/0	192.168.1.1	255.255.255.0	—
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.20	255.255.255.0	192.168.1.1
Router0	G0/1	192.168.2.1	255.255.255.0	—
PC2	NIC	192.168.2.10	255.255.255.0	192.168.2.1

Device required

Device	Quantity
Cisco 2911 Router	1
Cisco 2960 Switch	2
PC-PT	3
Copper Straight-Through Cable	5

Configure Router0
Open:
Router0 → CLI
Enter:
enable
configure terminal

Step 1 – Configure G0/0
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

Step 2 – Configure G0/1
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

Step 3: Configure PC0
Go to:
PC0 → Desktop → IP Configuration
Enter:
IP Address:
192.168.1.10

Subnet Mask:

255.255.255.0

Default Gateway:
192.168.1.1

Step 4 :Configure PC1
Go to:
PC1 → Desktop → IP Configuration
Enter:
IP Address:
192.168.1.20

Subnet Mask:
255.255.255.0

Default Gateway:
192.168.1.1

Step 5: Configure PC2
Go to:
PC2 → Desktop → IP Configuration
Enter:
IP Address:
192.168.2.10

Subnet Mask:
255.255.255.0

Default Gateway:
192.168.2.1

Step 6: Test the Network Before Applying ACL

Before configuring the ACL, make sure that all PCs can communicate.
Test PC0 → PC2
On PC0:
Desktop → Command Prompt
ping 192.168.2.10
Expected:
Reply from 192.168.2.10

Test PC1 → PC2
On PC1:
ping 192.168.2.10
Expected:
Reply from 192.168.2.10
At this stage:

PC0 → PC2 ALLOWED
PC1 → PC2 ALLOWED
This confirms that the basic network is working before the ACL is introduced.

Step 7: Create the ACL

Now we will create an ACL that blocks:

192.168.1.20 which is the address of **PC1**.

The ACL will allow all other source addresses.

Step 8:Configure Standard ACL

On Router0:
enable
configure terminal
Enter:
access-list 10 deny host 192.168.1.20
This means:

Deny traffic originating from PC1.

Now enter:
access-list 10 permit any
This means:

Permit all other source addresses.

Step 9: Apply the ACL to the Interface

We want to restrict access **towards the second LAN**.
Therefore, apply the ACL to the interface connected to LAN 2.
That is:
G0/1
We will apply it in the **outbound direction**.
Enter:
interface gigabitEthernet 0/1
ip access-group 10 out
exit
Then:
end
Save:
copy running-config startup-config

Step 10: Test the ACL
Test 1 – PC1 → PC2
From PC1:
ping 192.168.2.10
The ping should now **fail**.
You may see:
Request timed out.
or:

Destination host unreachable.
The important result is:
PC1 → PC2 = BLOCKED

Test PC0 → PC2

Now go to PC0:
ping 192.168.2.10
This should still succeed.
PC0 → PC2 = ALLOWED
This demonstrates that the ACL is blocking **only PC1**, not the entire LAN.

Test PC0 → PC1

PC0 and PC1 are on the same LAN:
192.168.1.0/24
Try:
ping 192.168.1.20
This should succeed.
Why?
Because the packet between PC0 and PC1 does not pass through Router0. Therefore, the ACL on Router0 G0/1 does not affect it.
This is an important observation.

Test PC1 → Router0

From PC1:
ping 192.168.1.1
This should normally succeed because the ACL was applied **outbound on G0/1**, not inbound on G0/0.

Step 11:Verify the ACL

On Router0:
show access-lists
You should see something similar to:
Standard IP access list 10
10 deny host 192.168.1.20
20 permit any
You may also see packet counters, for example:
10 deny host 192.168.1.20 (5 matches)
20 permit any (5 matches)
The match counts increase as traffic passes through the ACL.

Result

A standard IPv4 ACL was successfully configured on Router0 in Cisco Packet Tracer to restrict PC1 (**192.168.1.20**) from accessing PC2 (**192.168.2.10**). Access from PC0 was permitted, demonstrating selective traffic control using a standard ACL. The ACL operation was verified using **show access-lists**, **show ip interface**, and connectivity tests.

16. Blocking Web Access to a Specific Server Using an Extended ACL

Problem Statement

You are the network administrator of a college. The college is assigned the network address:**140.80.0.0** The college has **20 subnets**. The **Central Computing Facility (CCF)** is located in the **4th subnet**. The CSE department's hackathon registration website is hosted on a server assigned the **7th host address in the 16th subnet**. The registration has now closed. Therefore, students in the CCF must be prevented from accessing the hackathon website.

Aim

To design and configure a subnetted college network in Cisco Packet Tracer and use an **extended IPv4 ACL** to block **HTTP/HTTPS access** from the CCF subnet to a specific web server while allowing other services to remain accessible.

Background Theory

The college has been assigned:
140.80.0.0
Since 140 belongs to the Class B range, the default network mask is:
255.255.0.0
or:
/16
We need at least **20 subnets**.
To create 20 or more subnets:
$2^n \geq 20$
The smallest value of n satisfying this is:
$2^5 = 32$
Therefore, we borrow **5 bits** from the host portion.
Original:
/16
After borrowing 5 bits:
/21
Therefore, the subnet mask is:
255.255.248.0
This gives:
$2^5 = 32$ subnets
which is sufficient for the required 20 subnets.

4. Subnet Calculation

The subnet mask is:

255.255.248.0
The interesting octet is the **third octet**.

The block size is:

256 - 248 = 8
Therefore, the subnet addresses are:

Subnet No.	Network Address	Usable Host Range
1	140.80.0.0/21	140.80.0.1 – 140.80.7.254
2	140.80.8.0/21	140.80.8.1 – 140.80.15.254
3	140.80.16.0/21	140.80.16.1 – 140.80.23.254
4	**140.80.24.0/21**	**140.80.24.1 – 140.80.31.254**
5	140.80.32.0/21	140.80.32.1 – 140.80.39.254
6	140.80.40.0/21	140.80.40.1 – 140.80.47.254
7	140.80.48.0/21	140.80.48.1 – 140.80.55.254
8	140.80.56.0/21	140.80.56.1 – 140.80.63.254
9	140.80.64.0/21	140.80.64.1 – 140.80.71.254
10	140.80.72.0/21	140.80.72.1 – 140.80.79.254
11	140.80.80.0/21	140.80.80.1 – 140.80.87.254
12	140.80.88.0/21	140.80.88.1 – 140.80.95.254
13	140.80.96.0/21	140.80.96.1 – 140.80.103.254
14	140.80.104.0/21	140.80.104.1 – 140.80.111.254
15	140.80.112.0/21	140.80.112.1 – 140.80.119.254
16	**140.80.120.0/21**	**140.80.120.1 – 140.80.127.254**
17	140.80.128.0/21	140.80.128.1 – 140.80.135.254
18	140.80.136.0/21	140.80.136.1 – 140.80.143.254
19	140.80.144.0/21	140.80.144.1 – 140.80.151.254
20	140.80.152.0/21	140.80.152.1 – 140.80.159.254

Identify the CCF Network

The CCF is in the **4th subnet**.
Therefore:
CCF Network = 140.80.24.0/21
Usable addresses:
140.80.24.1
to
140.80.31.254
We will use:
Router CCF interface = 140.80.24.1

and assign PCs addresses such as:
PC0 = 140.80.24.10
PC1 = 140.80.24.11
PC2 = 140.80.24.12

Identify the Hackathon Server Network

The server is in the **16th subnet**.

Therefore:

16th subnet = 140.80.120.0/21
Usable range:

140.80.120.1
to
140.80.127.254
The problem states that the server is assigned the **7th address**.

For this experiment, we interpret "7th address" as the **7th usable host address**:

140.80.120.7
Therefore:

Hackathon Server IP = 140.80.120.7

Final Addressing Scheme

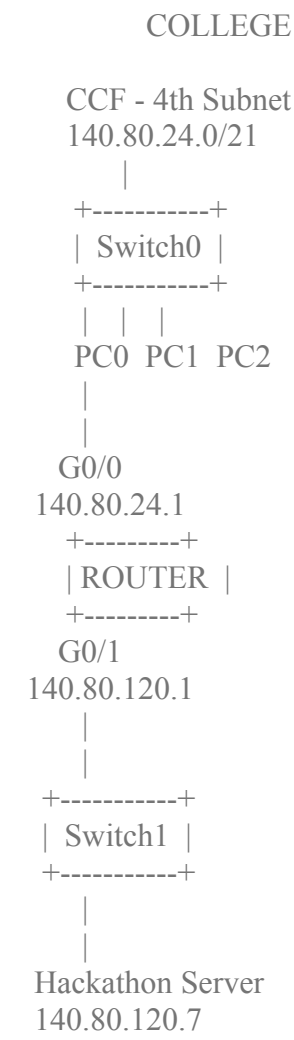
Device	Interface	IP Address	Subnet Mask
Router	G0/0	140.80.24.1	255.255.248.0
Router	G0/1	140.80.120.1	255.255.248.0
CCF PC0	Fa0	140.80.24.10	255.255.248.0
CCF PC1	Fa0	140.80.24.11	255.255.248.0
CCF PC2	Fa0	140.80.24.12	255.255.248.0
Hackathon Server	Fa0	140.80.120.7	255.255.248.0

Default gateways:

CCF PCs → 140.80.24.1
Server → 140.80.120.1

Network Topology

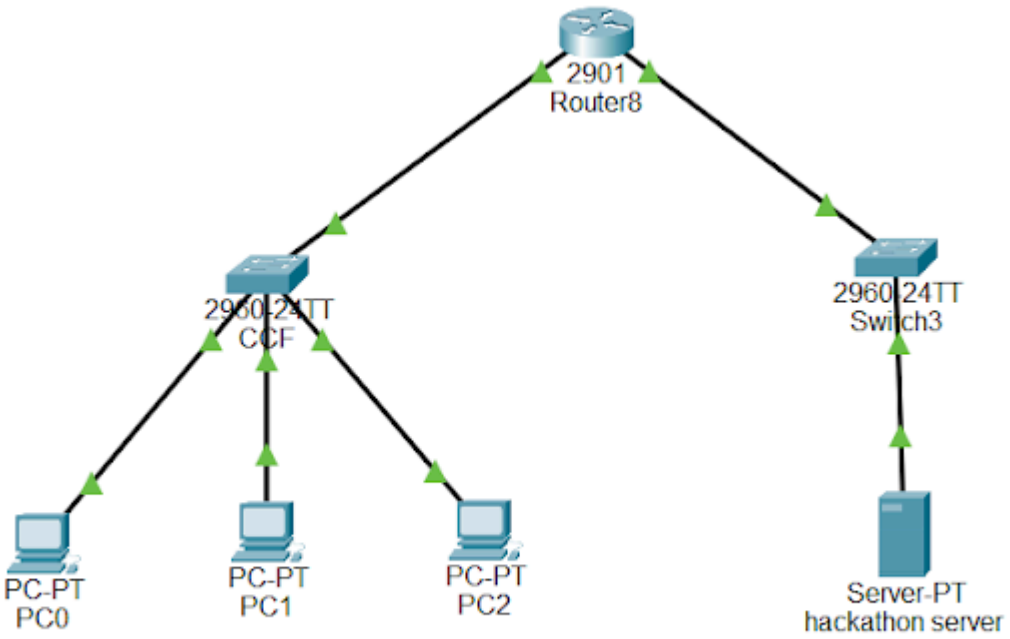
For Packet Tracer, we can represent the required networks with two LANs connected by a router.



The router connects:

CCF network
140.80.24.0/21
to:

Server network
140.80.120.0/21



Devices Required

Device	Quantity
Cisco Router	1
Cisco 2960 Switch	2
PCs	3
Server	1
Copper Straight-Through Cables	5

An Extended ACL can examine:

Source IP
Destination IP
Protocol
Source port
Destination port

Therefore, we can specifically say:

CCF → Hackathon Server → TCP → port 80

and:

CCF → Hackathon Server → TCP → port 443

should be denied.

Everything else can be permitted.

Ports Used by Web Services

Service	Protocol	Port
HTTP	TCP	80
HTTPS	TCP	443
FTP	TCP	21
DNS	UDP/TCP	53
SSH	TCP	22
Telnet	TCP	23

Therefore, to block only web access, we need to block:

TCP port 80
TCP port 443
while allowing other traffic.

Procedure

Task 1: Open Cisco Packet Tracer

1. Open Cisco Packet Tracer.
2. Create a new workspace.

Task 2: Place the Router

1. Select **Network Devices → Routers**.
2. Select a router with at least two Ethernet/GigabitEthernet interfaces.
3. Place the router in the workspace.

For example:

Router0

Task 3: Place Two Switches

Place two 2960 switches.

Switch0 → CCF LAN
Switch1 → Server LAN

Task 4: Place Three PCs

Place:

PC0
PC1
PC2
These represent students accessing the network from CCF.

Task 5: Place the Server

From **End Devices**, place one server.

Rename it conceptually:

Hackathon-Web-Server

Task 6: Connect the CCF LAN

Use Copper Straight-Through cables.

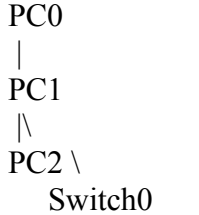
Connect:

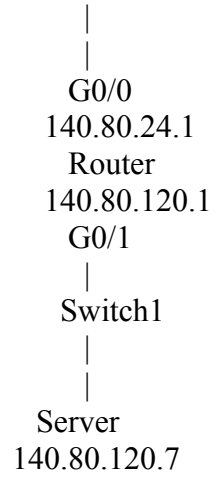
PC0 → Switch0
PC1 → Switch0
PC2 → Switch0
Switch0 → Router G0/0

Task 7: Connect the Server LAN

Connect:

Router G0/1 → Switch1
Switch1 → Server
The topology should look like:





---

13. Configure the Router

Task 8: Configure G0/0

Open:

Router → CLI

Enter:

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 140.80.24.1 255.255.248.0
no shutdown
exit
```

Task 9: Configure G0/1

Enter:

```
interface gigabitEthernet 0/1
ip address 140.80.120.1 255.255.248.0
no shutdown
exit
end
```

Verify Router Interfaces

Task 10

Enter:

```
show ip interface brief
```

Expected:

Interface	IP Address	Status	Protocol
G0/0	140.80.24.1	up	up
G0/1	140.80.120.1	up	up

Both interfaces should ideally show:

up
up

Configure CCF PCs

Task 11: Configure PC0

Go to:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 140.80.24.10
Subnet Mask: 255.255.248.0
Default Gateway: 140.80.24.1

Task 12: Configure PC1

IP Address: 140.80.24.11
Subnet Mask: 255.255.248.0
Default Gateway: 140.80.24.1

Task 13: Configure PC2

IP Address: 140.80.24.12
Subnet Mask: 255.255.248.0
Default Gateway: 140.80.24.1

Configure the Hackathon Server

Task 14

Go to:

Server → Desktop → IP Configuration

Enter:

IP Address: 140.80.120.7
Subnet Mask: 255.255.248.0

Default Gateway: 140.80.120.1

Configure the Web Server

Task 15: Enable HTTP

Go to:

Server → Services → HTTP

Make sure:

HTTP: ON
You can edit the default web page if desired.

For example, use:

HACKATHON REGISTRATION

Registration is now CLOSED.
This makes the experiment visually clear when students access the website.

Test Connectivity Before Applying ACL

Task 16: Ping the Router

From PC0:

ping 140.80.24.1
Expected:

Successful

Task 17: Ping the Server

From PC0:

ping 140.80.120.7
Expected:

Successful

Do the same from PC1:

ping 140.80.120.7
and PC2:

ping 140.80.120.7
All should succeed.

Test the Website Before Applying ACL

Task 18

On PC0:

Desktop → Web Browser

Enter:

http://140.80.120.7
The server's web page should open.

Repeat from PC1 and PC2.

Therefore, before the ACL:

CCF → HTTP → Server is working.

Important Observation Before ACL

Test	Expected Result
PC0 → Router ping	Success
PC0 → Server ping	Success
PC1 → Server ping	Success
PC2 → Server ping	Success
PC0 → HTTP website	Success
PC1 → HTTP website	Success
PC2 → HTTP website	Success

This establishes the **baseline**.

Configure the Extended ACL

Now we implement the actual requirement.

We will create an ACL called:

CCF_WEB_BLOCK
The ACL will:

1. Deny HTTP from CCF to the server.
2. Deny HTTPS from CCF to the server.
3. Permit everything else.

Enter ACL Configuration Mode

On the router:

enable
configure terminal
Create the extended named ACL:

ip access-list extended CCF_WEB_BLOCK
The prompt becomes:

Router(config-ext-nacl)#

Task 20: Block HTTP

Enter:

deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 80
Let's understand this command:

deny
Block the traffic.

tcp
Only TCP traffic.

140.80.24.0 0.0.7.255
Source = entire CCF subnet.

host 140.80.120.7
Destination = only the hackathon server.

eq 80
Destination TCP port 80 = HTTP.

Task 21: Block HTTPS

Enter:

deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443
This blocks HTTPS.

Task 22: Permit All Other Traffic

This line is **very important**.

Enter:

permit ip any any
Why?

Because ACLs have an implicit:

deny ip any any
at the end.

If we don't explicitly permit other traffic, the ACL could unintentionally block other communications.

Therefore:

deny HTTP
deny HTTPS
permit everything else is exactly what we need.

Complete ACL Configuration

The complete configuration is:

Router(config)# ip access-list extended CCF_WEB_BLOCK

Router(config-ext-nacl)# deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 80

Router(config-ext-nacl)# deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443

Router(config-ext-nacl)# permit ip any any

Router(config-ext-nacl)# exit

Task 23: Apply the ACL to the Correct Interface

Now we have to decide where to apply the ACL.

We want to filter traffic coming from CCF.

Therefore, we can apply the ACL inbound on G0/0, the interface connected to CCF.

Enter:

interface gigabitEthernet 0/0
ip access-group CCF_WEB_BLOCK in
exit
end
The complete command is:

Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip access-group CCF_WEB_BLOCK in
Router(config-if)# exit
Router(config)# end

Test the ACL

Now repeat the tests.

Task 24: Ping the Server

From PC0:

ping 140.80.120.7

Expected:
SUCCESS

This is very important.

The ping should still work because:

ping = ICMP

and we only blocked:

TCP port 80
TCP port 443

Test HTTP Access

Task 25

From PC0:

Desktop → Web Browser

Enter:

http://140.80.120.7

Expected result:
Website access should fail.

Repeat from:

PC1
PC2
The website should be inaccessible from all CCF PCs.

Test HTTPS Access

If HTTPS is enabled:

https://140.80.120.7

Expected:
Blocked

This verifies that both web protocols are blocked.

Test ICMP

From PC0:

ping 140.80.120.7
Expected:

SUCCESS

Therefore:

ICMP → Allowed

Test FTP

If you enable FTP on the server:

Go to:

Server → Services → FTP

Set:

FTP: ON
Create a test FTP user if required by your Packet Tracer version.

Then from PC0, use:

Desktop → Command Prompt

Try:

ftp 140.80.120.7
Expected:

FTP access should continue to work.

This demonstrates that:

TCP port 21
has not been blocked.

Verify ACL Statistics

On the router:

show access-lists
You should see something similar to:

Extended IP access list CCF_WEB_BLOCK

deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq www
deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443
permit ip any any
The ACL entries may display hit counts.

For example:

deny tcp ... eq 80 (5 matches)
deny tcp ... eq 443 (2 matches)
permit ip any any (15 matches)
The exact counts will depend on how many tests were performed.

Task 26: Observe ACL Hit Counts

Perform:

show access-lists
before and after attempting to access:

http://140.80.120.7
Students should observe that the match count of the HTTP deny rule increases.

This provides evidence that the ACL is actually processing the web traffic.

Complete ACL Configuration

Students should record this in their lab record:

```
enable
configure terminal

ip access-list extended CCF_WEB_BLOCK

deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 80

deny tcp 140.80.24.0 0.0.7.255 host 140.80.120.7 eq 443

permit ip any any

exit

interface gigabitEthernet 0/0

ip access-group CCF_WEB_BLOCK in

exit
end
```

Result

The college network was subnetted using a /21 subnet mask to create 32 subnets. The CCF was identified as the 4th subnet, 140.80.24.0/21, and the hackathon server was assigned the 7th usable host address of the 16th subnet, 140.80.120.7. An extended ACL was configured on the router to block HTTP and HTTPS traffic from the CCF subnet to the hackathon server. Other services, including ICMP and FTP, were permitted, thereby satisfying the requirement to block only website access.

17.IPv6 Dynamic Routing Using RIPng

Aim

To configure and implement **RIPng (Routing Information Protocol next generation)** for dynamic routing between two IPv6 networks using Cisco Packet Tracer and verify end-to-end IPv6 connectivity.